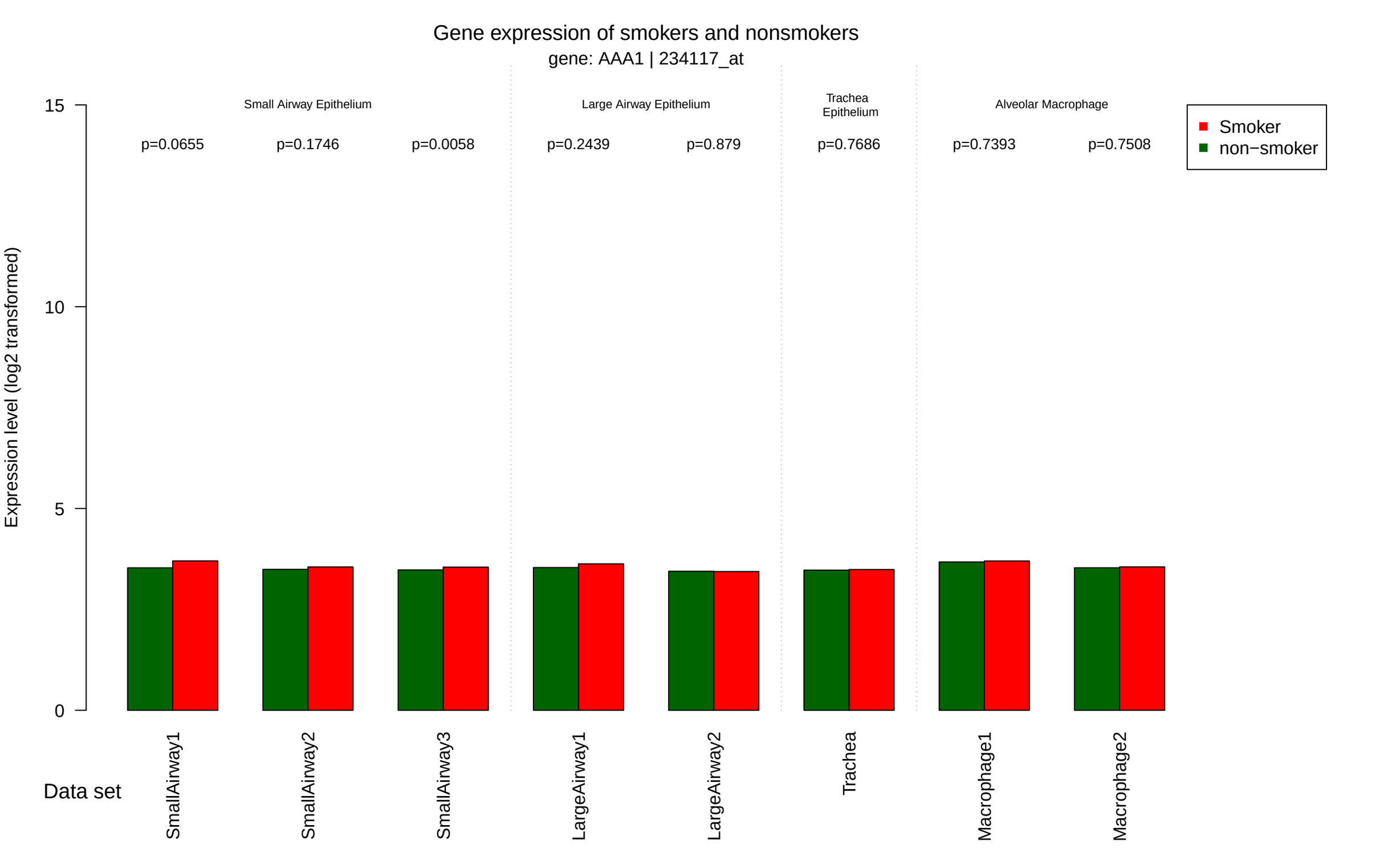
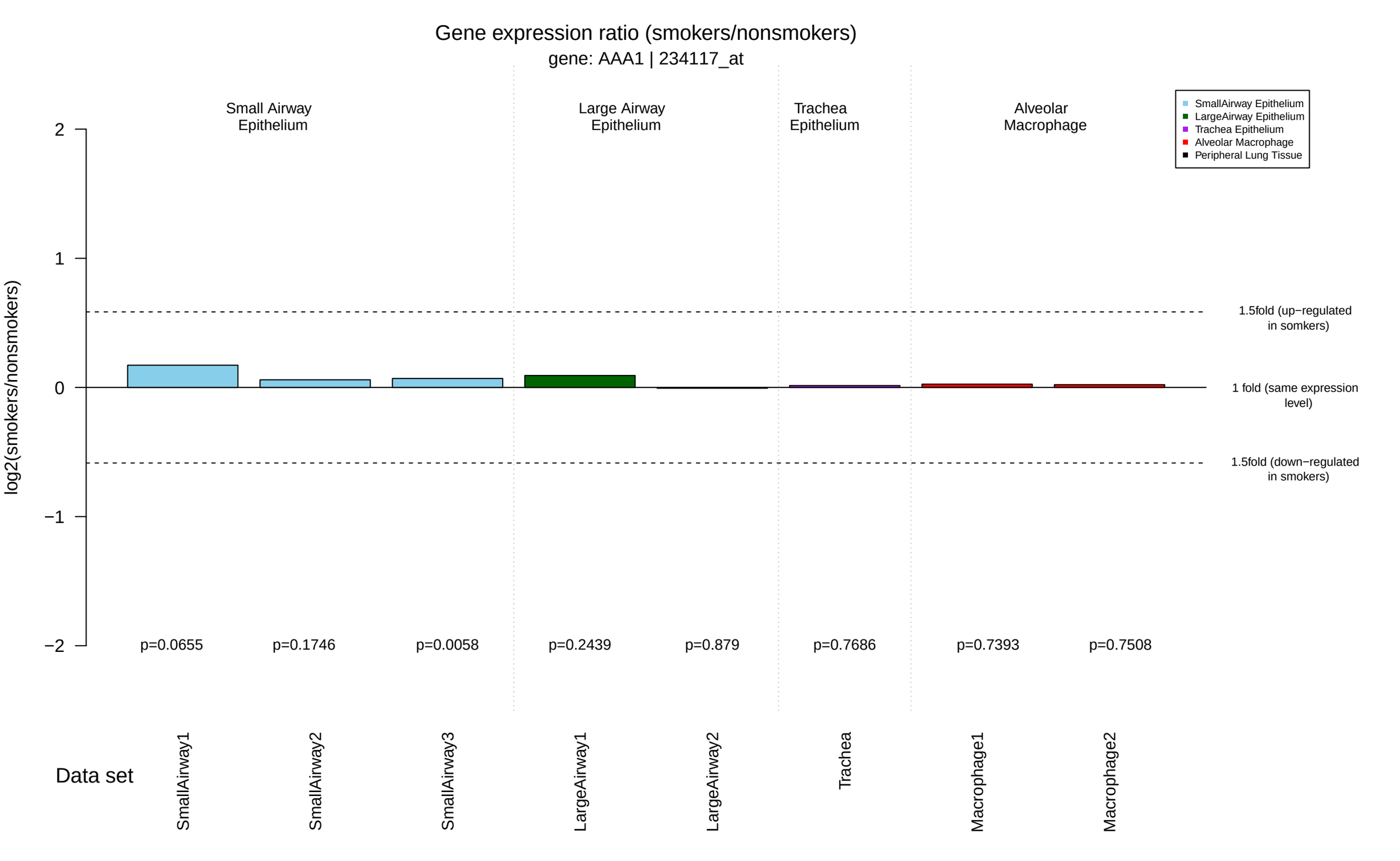
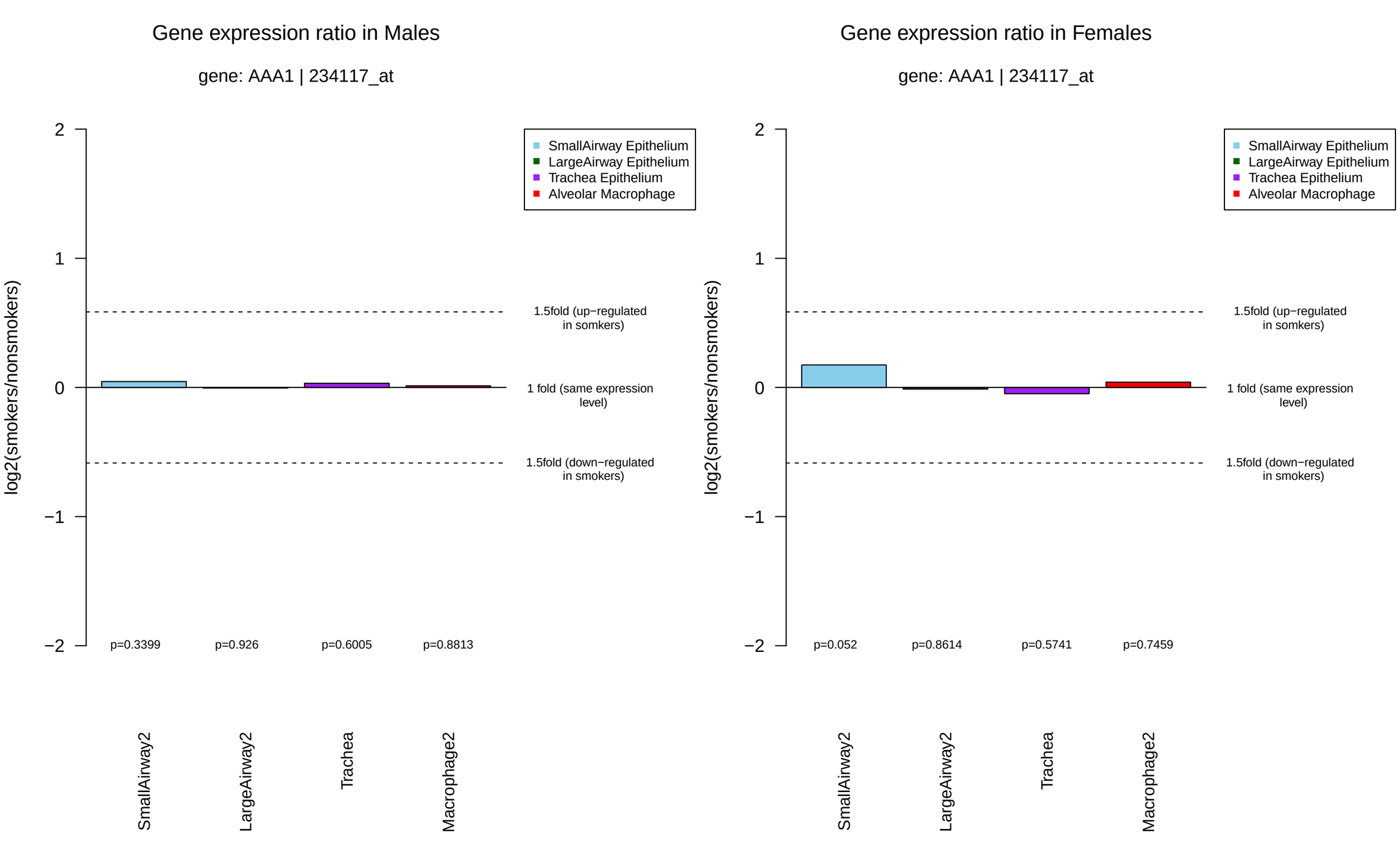
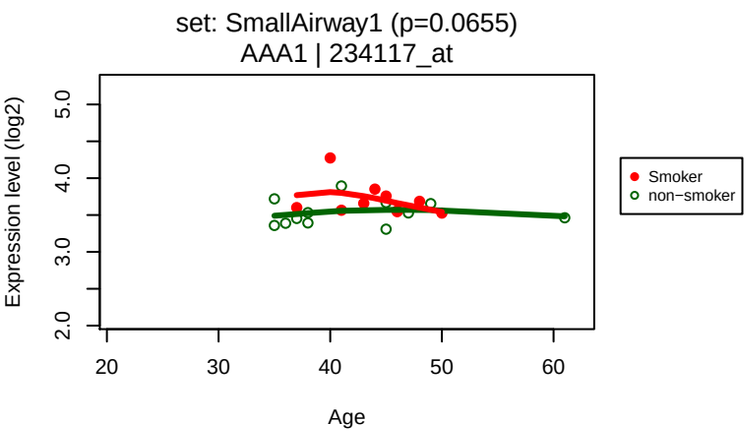
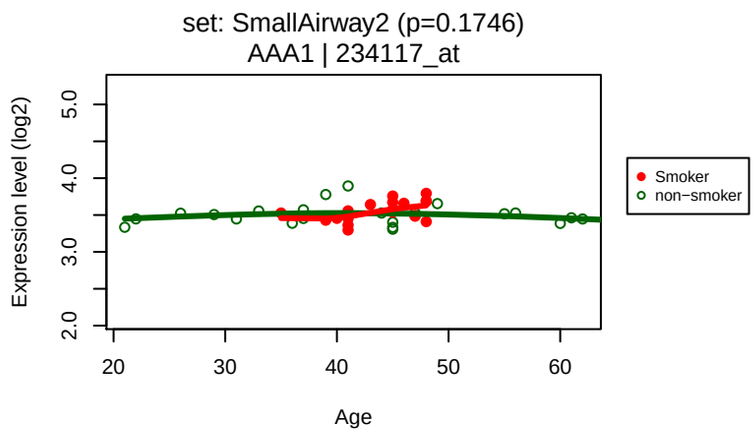
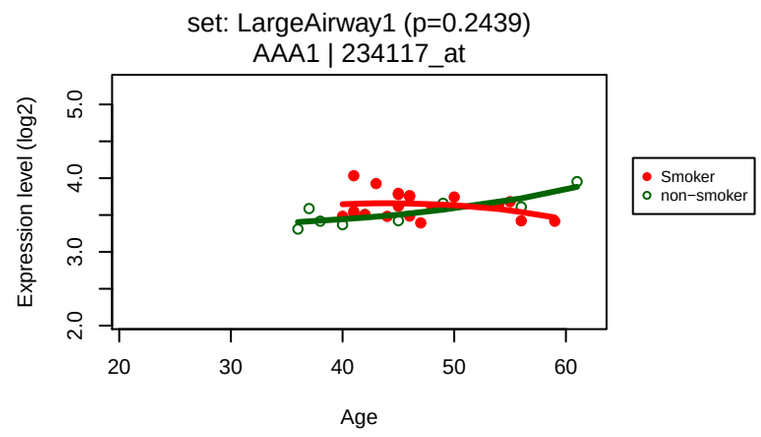

Gene expression of smokers and nonsmokers
gene: AAA1 | 234117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0655
p=0.1746
p=0.0058
p=0.2439
p=0.879
p=0.7686
p=0.7393
p=0.7508
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

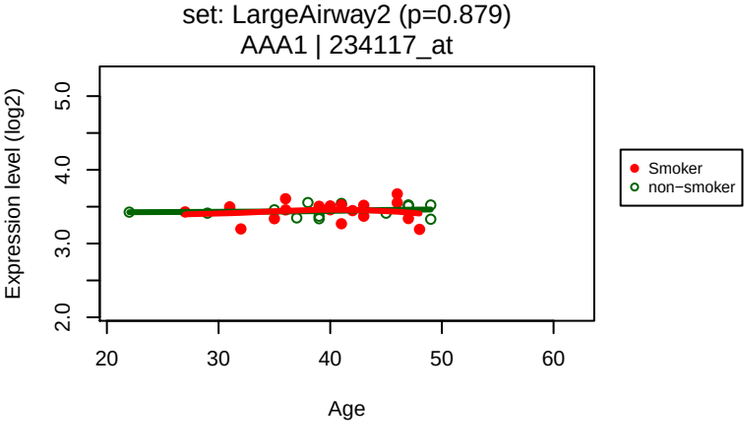

Gene expression ratio (smokers/nonsmokers)
gene: AAA1 | 234117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0655
p=0.1746
p=0.0058
p=0.2439
p=0.879
p=0.7686
p=0.7393
p=0.7508
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

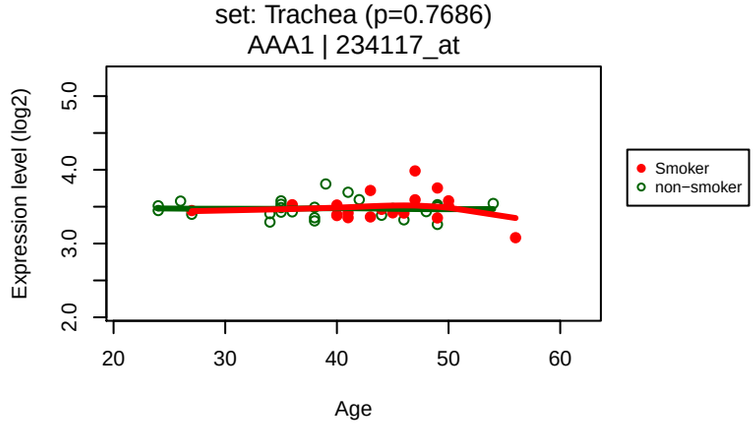

Gene expression ratio in Males
gene: AAA1 | 234117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3399
p=0.926
p=0.6005
p=0.8813
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AAA1 | 234117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.052
p=0.8614
p=0.5741
p=0.7459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

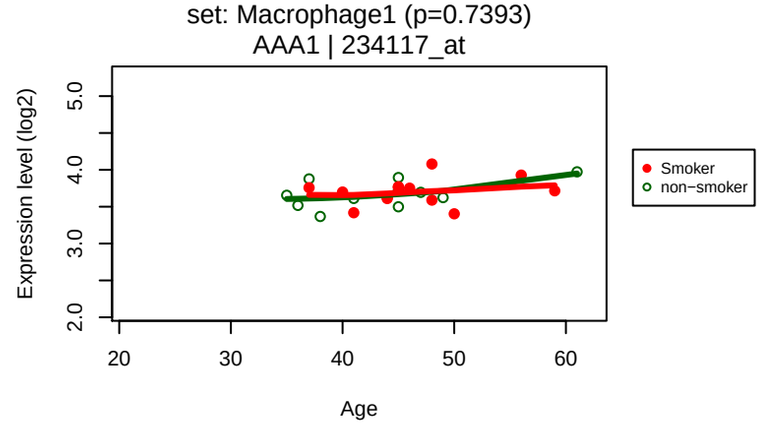

set: SmallAirway1 (p=0.0655)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker

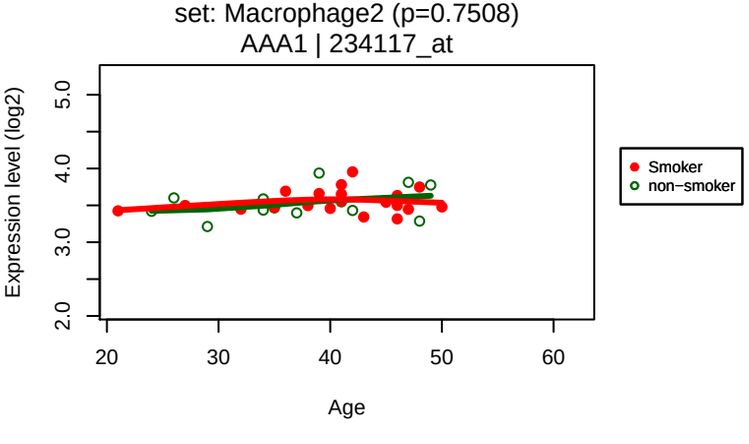

set: SmallAirway2 (p=0.1746)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker

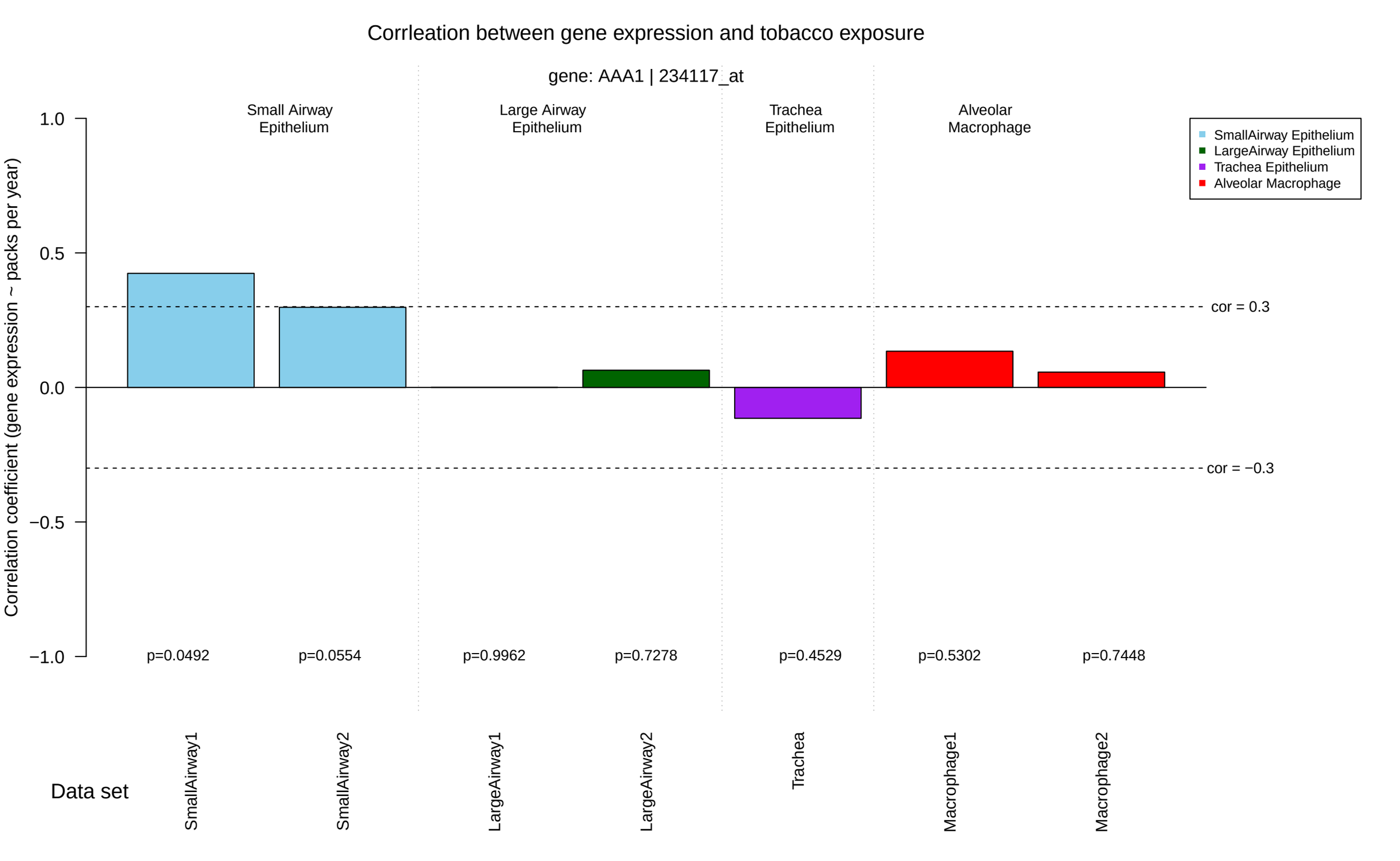

set: LargeAirway1 (p=0.2439)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.879)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7686)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7393)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7508)
AAA1 | 234117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AAA1 | 234117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0492
p=0.0554
p=0.9962
p=0.7278
p=0.4529
p=0.5302
p=0.7448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2